Process Model

ISYS3416/ISYS3432

Software Engineering Fundamentals for IT

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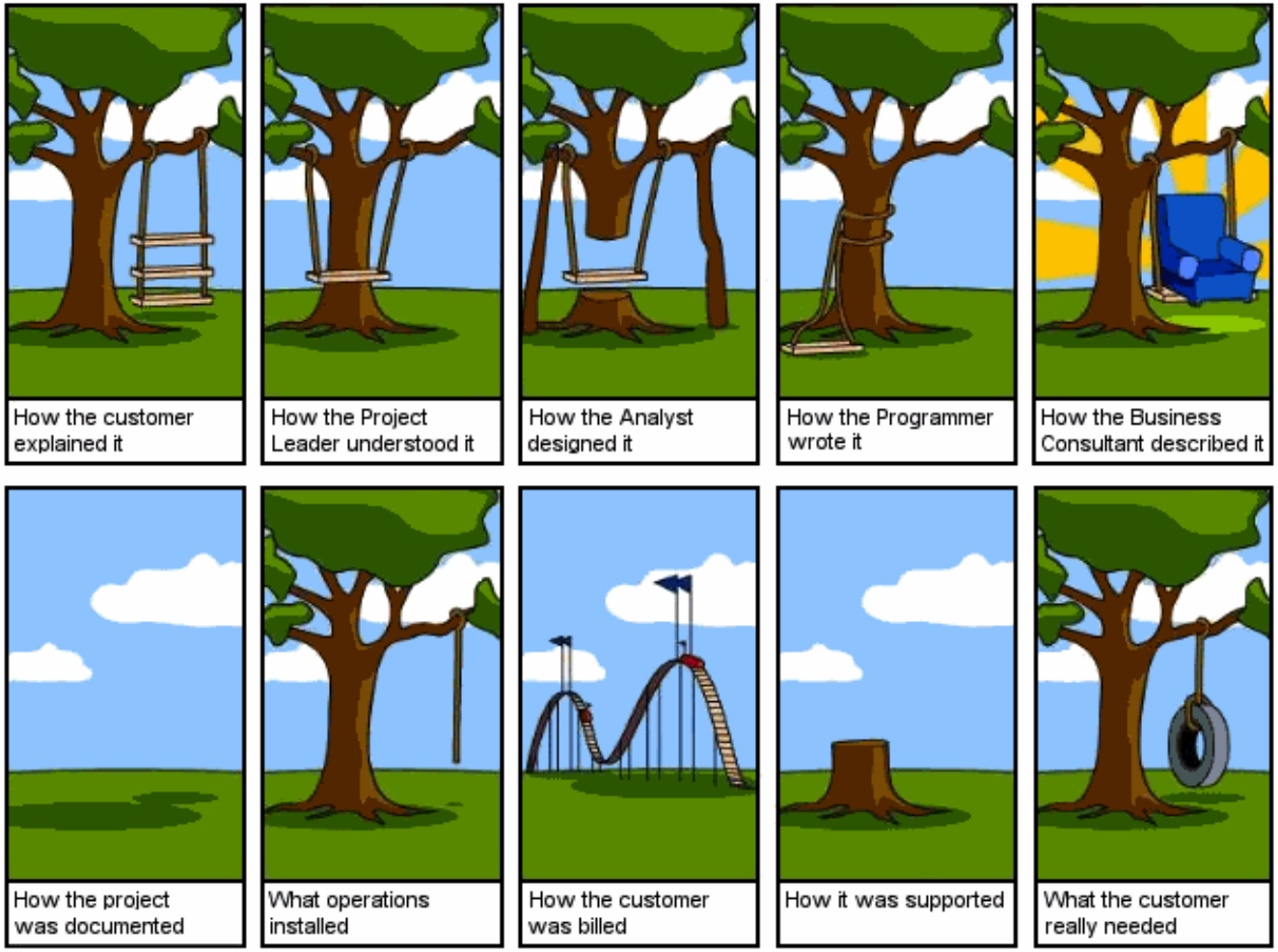
Module Learning Outcomes

Explain the fundamental concepts and characteristics of the major software process models

Compare and analyse different process models to determine their sustainability for various software development contexts

Apply and justify an appropriate process model for a given software project scenario

This is what will happen if you try to code a large project following this “process”…

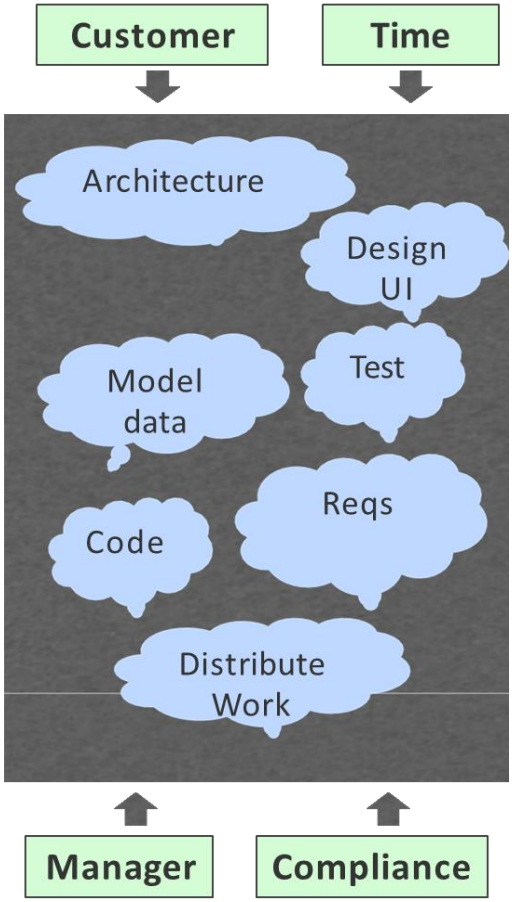


What is the Process Model

A process model is a structured collection of practices that describe the characteristics of effective process.

Many different software processes but all involve:

- Specification – defining what the system should do;
- Design and implementation – defining the organization of the system and implementing the system;
- Validation – checking that it does what the customer wants;
- Evolution – changing the system in response to changing customer needs



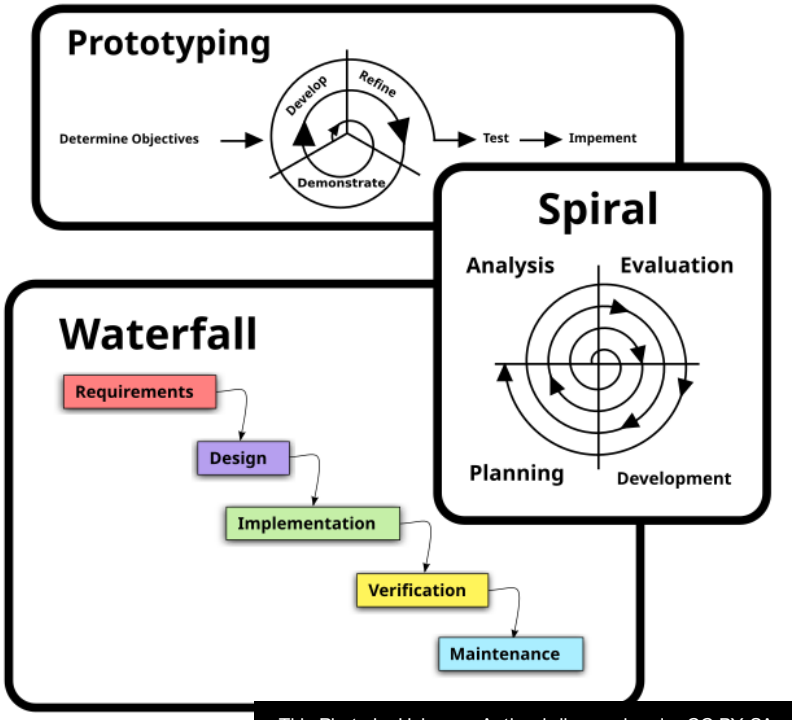
Software processes define the who, what, when, and how of developing software.

For example: A software process should specify the activities (*how*) a tester (*who*) should do, the order of these activities (*when*), and the artifacts (*what*) that should be produced by a tester.

How Is a Process Model Used?

A process model is used

- To help set process improvement objectives and priorities
- To help ensure stable, capable, and mature processes
- As a guide for improvement of project and organizational processes
- With an appraisal method to diagnose the state of an organization's current practices



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Why Is a Process Model Important?



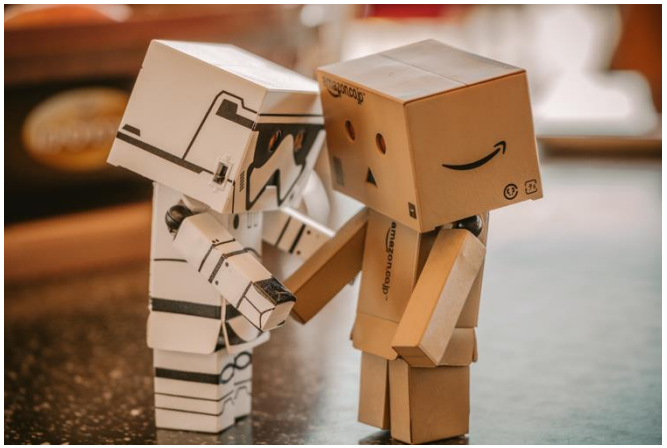
A process model provides

- A place to start improving
- The benefit of a community's prior experiences
- A common language and a shared vision
- A framework for prioritizing actions
- A way to define what improvement means for an organization

Plan-driven (heavy) vs Agile (light) Processes

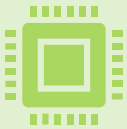
- In Plan-driven processes, all of the activities are planned in advance and progress is measured against this plan.
  Plan drives everything!
- In Agile Processes planning is incremental, and it is easier to change the process to reflect changing requirements.
- In practice, most practical processes include elements of both plan-driven and agile. There are no right or wrong software processes.
- What type of process would suit us for the assignment?

How can we understand the idea, communicate with each other, clarify what needs to be done, estimate how much time it will take and how much is it going to cost?



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Software Development Lifecycles (SDLC)



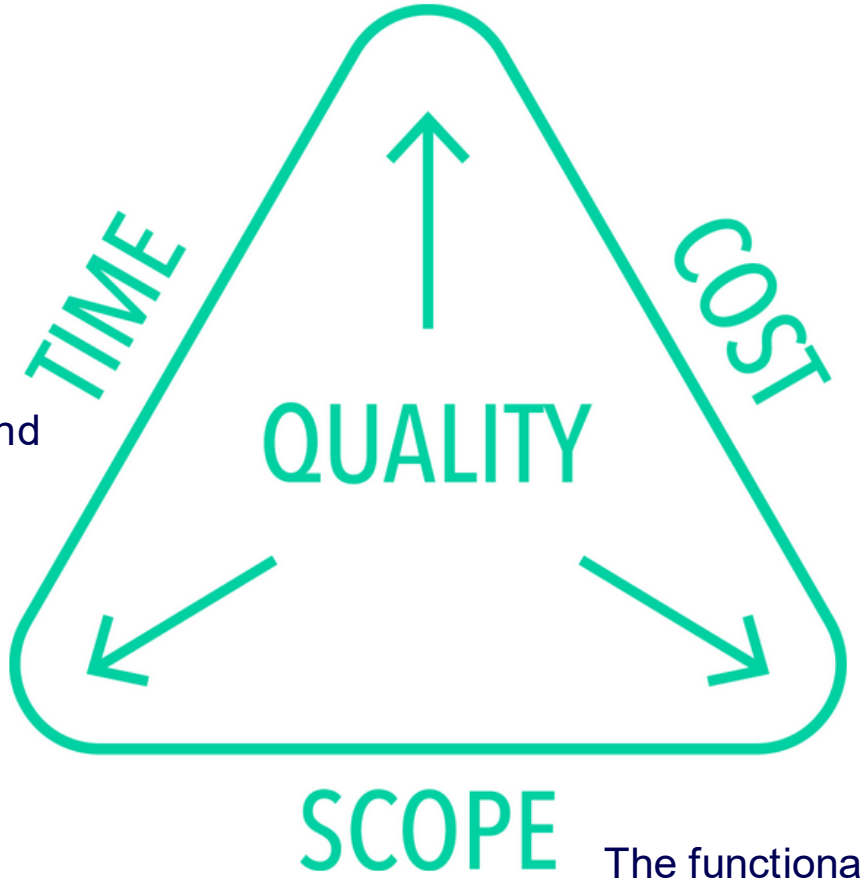
An SDLC is a process that produces software with the highest quality and lowest cost in the shortest time



It is composed of stages.

Project Management Triangle

The actual time required to produce a deliverable. Related to the amount of requirements and resources allocated

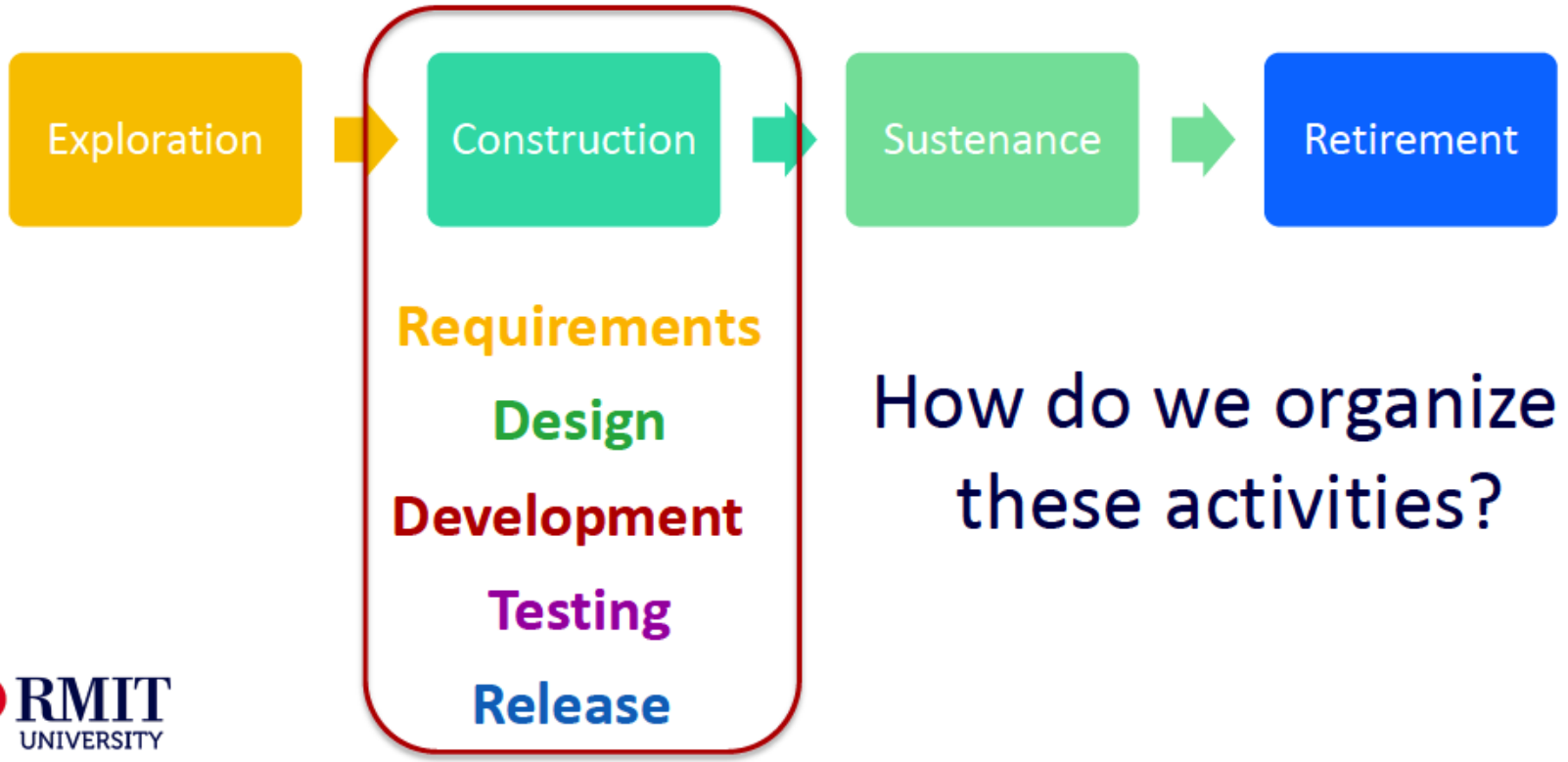


The amount of money needed to complete the project. Why does the time allocated affect the cost?

The functional elements that make up the project deliverables. The common success measure is its inherent quality upon delivery.

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General Project Stages

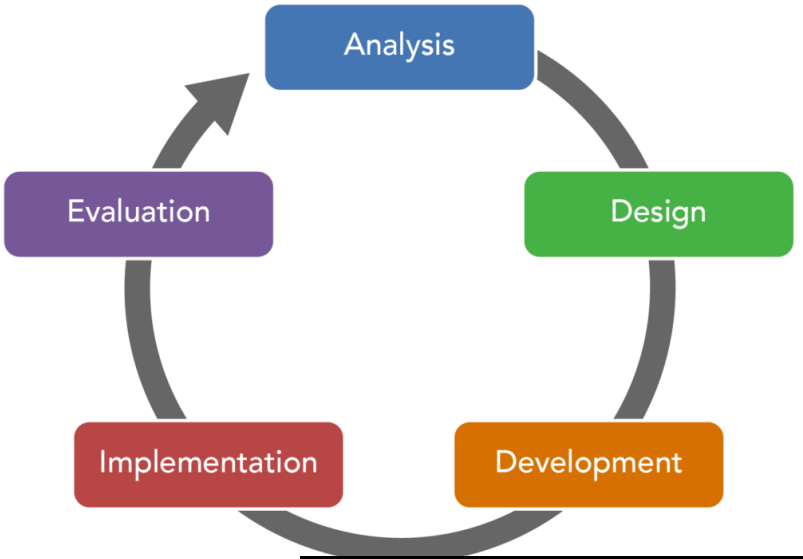


How do we organize these activities?

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Common Process Models

- Code and Fix Model
- Waterfall
- Rational Unified Process (RUP)
- Agile Development Cycle
  - Agile Unified Process (AUP)
  - eXtreme Programming (XP)
  - Scrum
  - Kanban
- Rapid Application Development (RAD)



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Code and Fix Model

Work from a brief specification of tasks

Developers immediately translate tasks into code.

Testing is carried out after all development work is completed

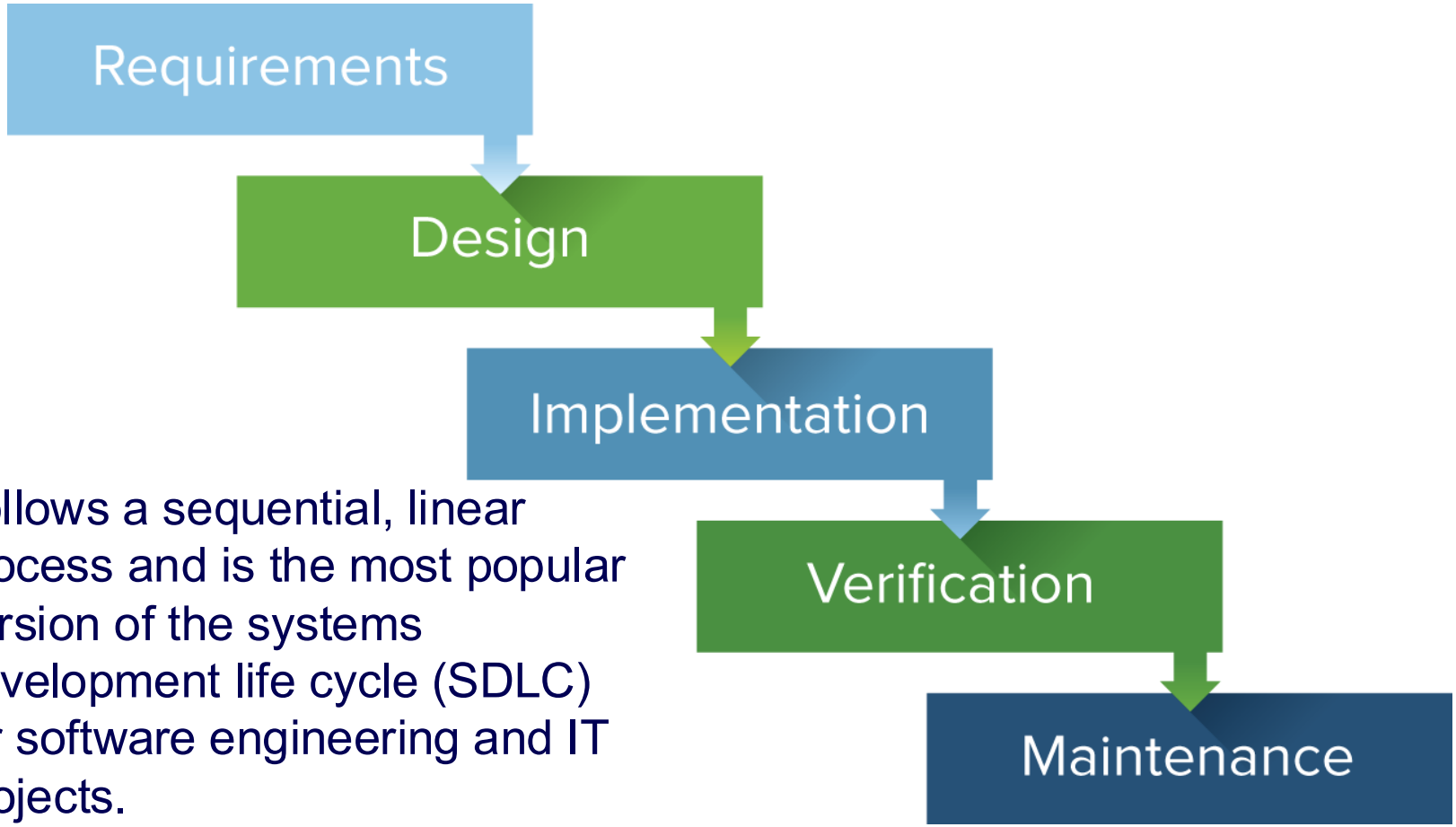
Developers would have no idea if they were producing what the customer wanted.

Approach worked well for scientific applications and batched programs where all development work is done by a single person.

But this approach did not scale well for larger applications

Need for better process models

Waterfall



Follows a sequential, linear process and is the most popular version of the systems development life cycle (SDLC) for software engineering and IT projects.

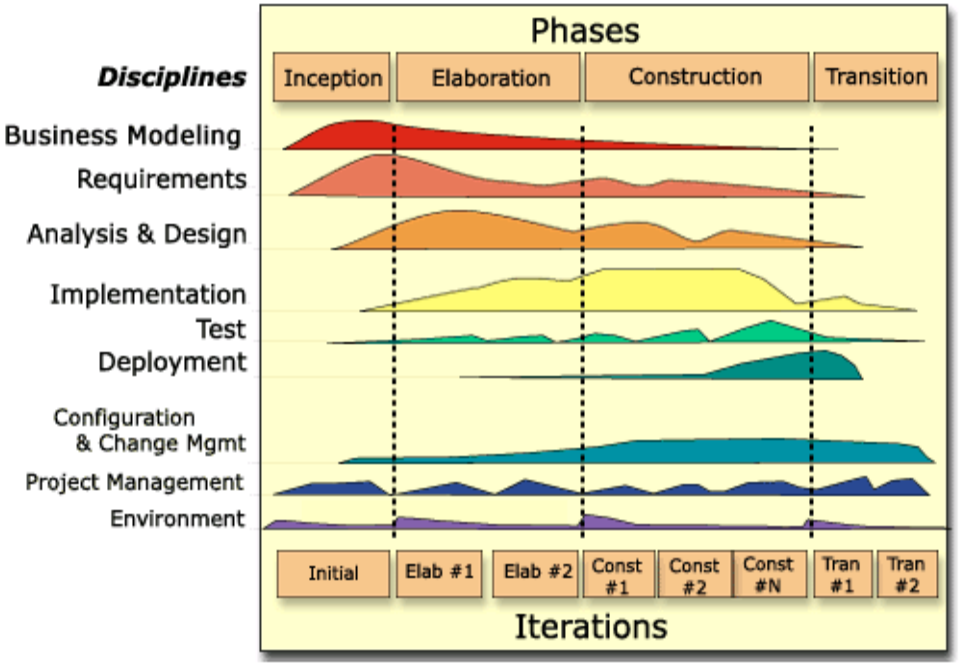
1) Requirements

Requirements are what the user needs and what the system needs to have.

Functional Requirements	Non Functional Requirements
• Behaviours we expect the system to have. • Functionalities that can be directly implemented. • There are several techniques to get these from the users.	• Also called *quality attributes*. • Other features that the system must conform (change screens fast enough, be pretty, be secure, do not take the whole CPU capacity). • Hard to capture, document, and get from the user.

Rational Unified Process (RUP)

- This methodology divides the development process into four distinct phases that each involves business modeling, analysis and design, implementation, testing, and deployment.
- This is an object-oriented and web-enabled program development methodology.
- This model also helps software developer for providing them guidelines, templates, and examples for all aspects and stages of software development.



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Agile Software Development

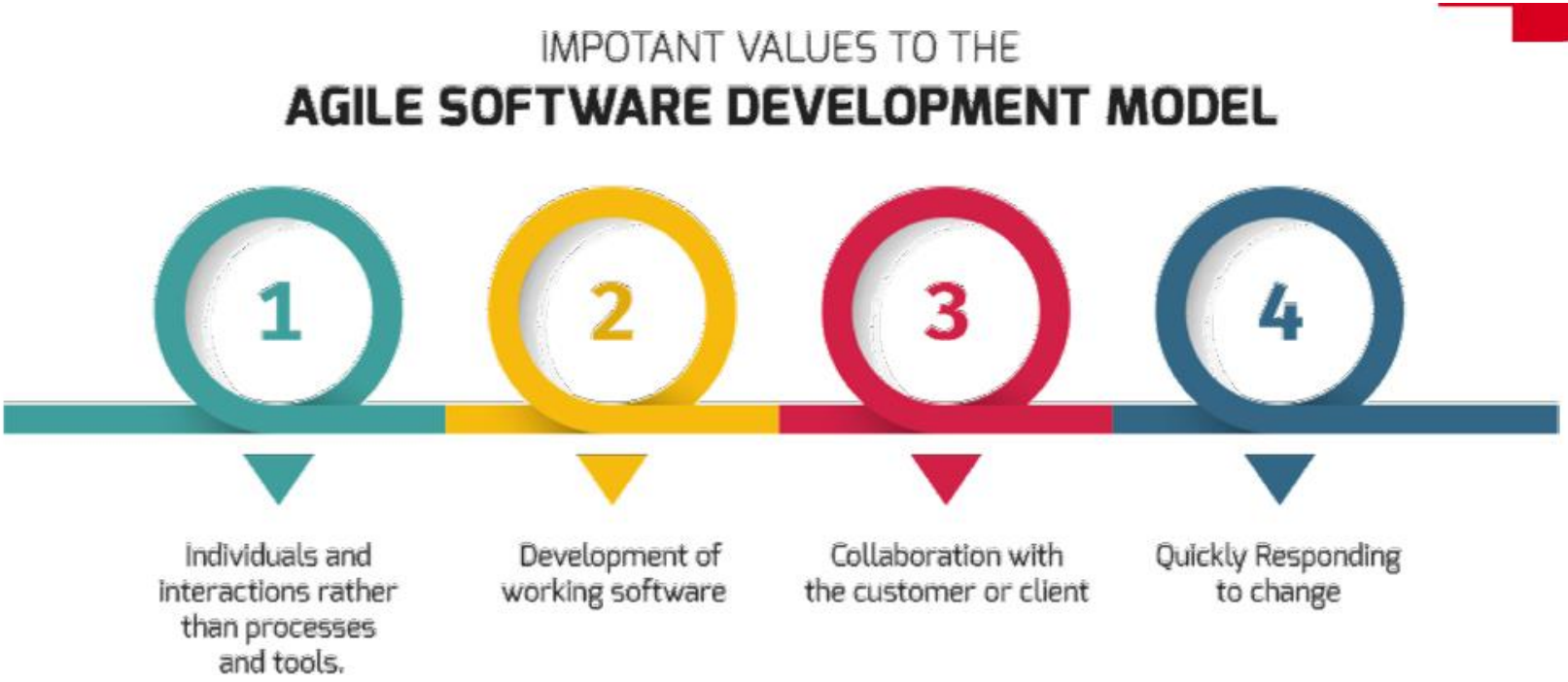
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What is Agile?

A set of principles for software development under which requirements and solutions evolve through the collaborative effort of self-organizing cross functional teams.

Is an iterative approach to software delivery that builds software incrementally from the start of the project, instead of trying to deliver it all at once near the end.

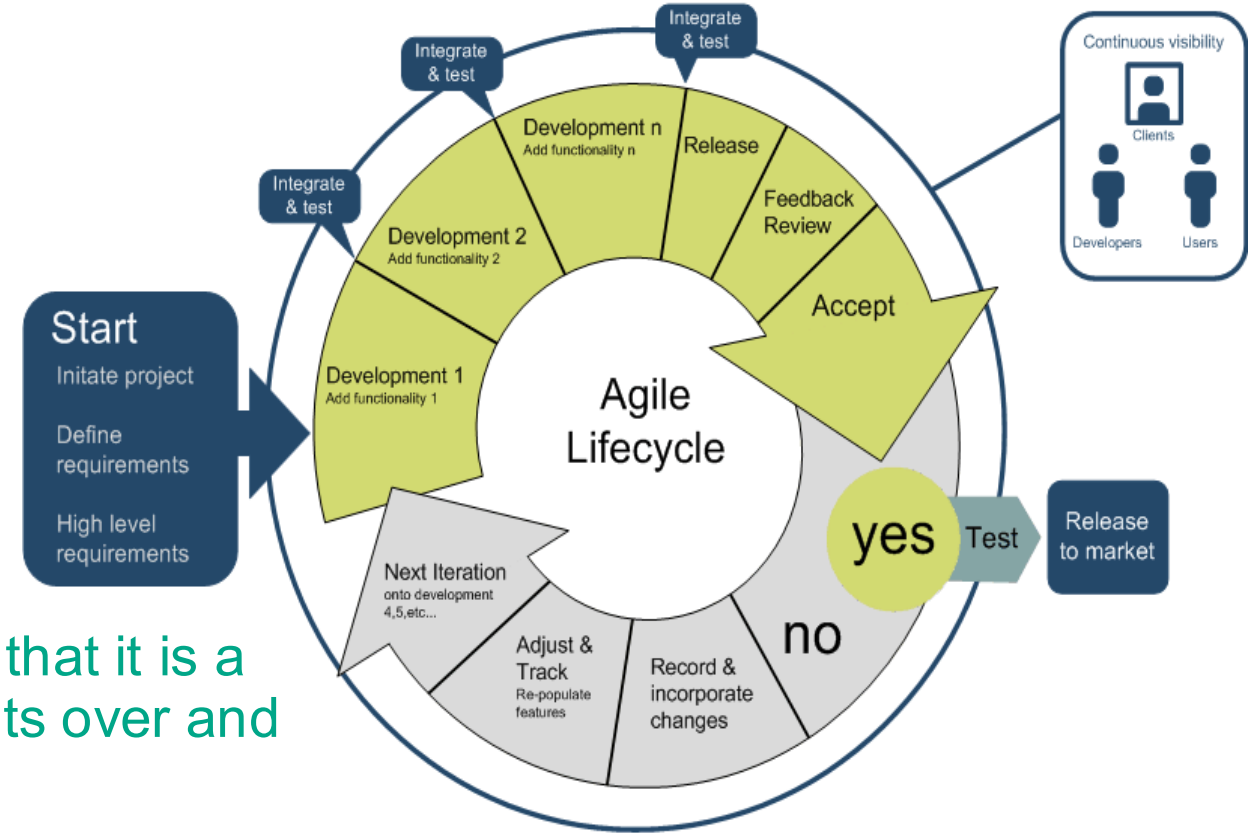
Agility Values



Agile is a type of methodology. There are tons of agile SDLC, some more used than others!

But in general, they all abide by the 4 values, and the 12 principles that are listed in the ***Agile Manifesto***.

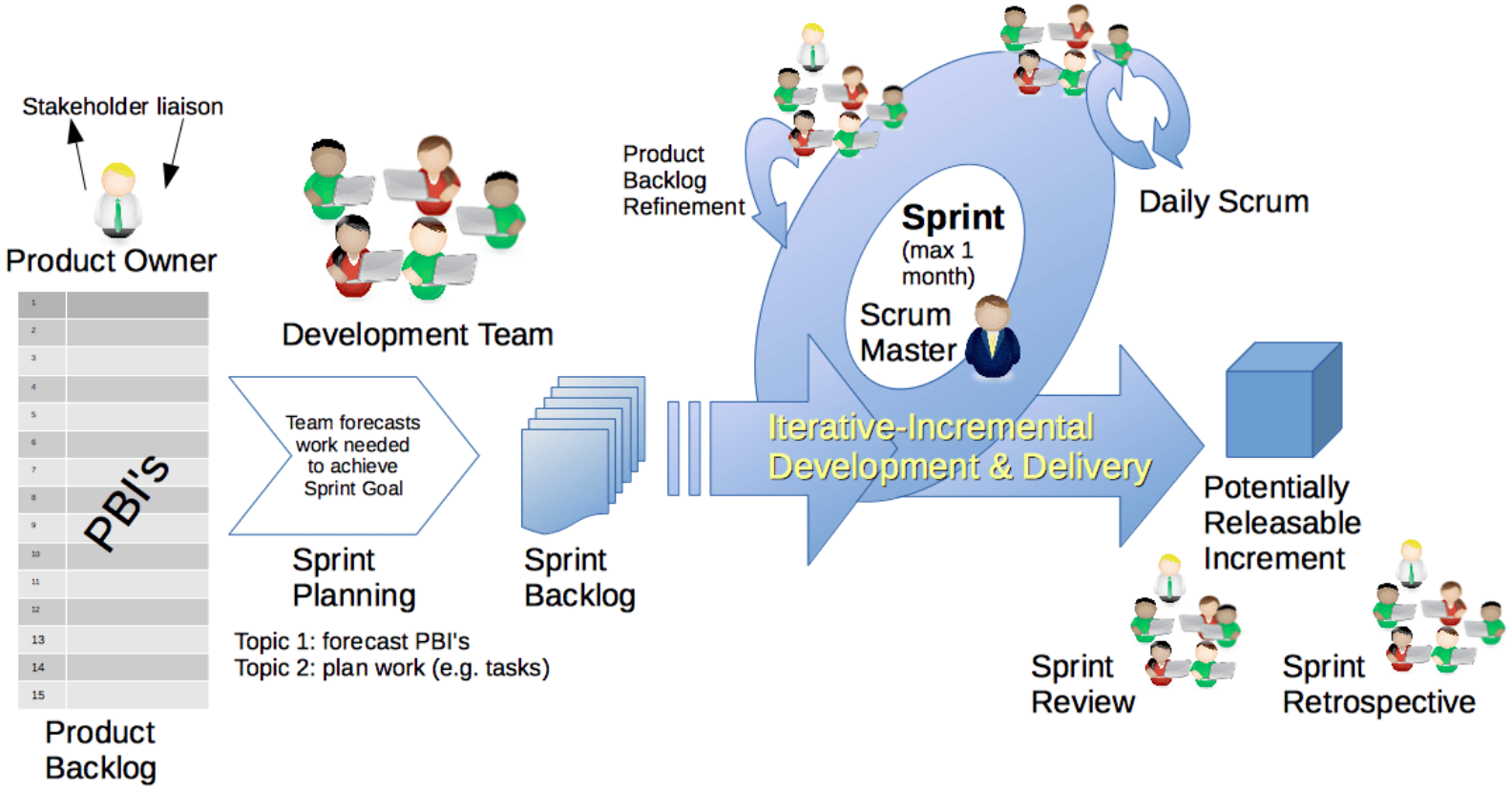
Agile SDLC are iterative and incremental



Iterative means that it is a cycle that repeats over and over.

Incremental means that you start with the basics, and each cycle (iteration!) builds on top of the previous one, making the system larger and more complex.

1) Scrum



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Backlogs

- The *product backlog* represents the larger picture and lists epics and stories for a product.

- A *sprint backlog* is concerned with the stories and tasks that are to be undertaken within a sprint. The items in sprint backlogs come from the product backlog during sprint planning.

- *Note: There is another term used at times called* Epic- a larger requirement or composite user story that needs to be broken down into small segments during software development.

Requirements with User Stories

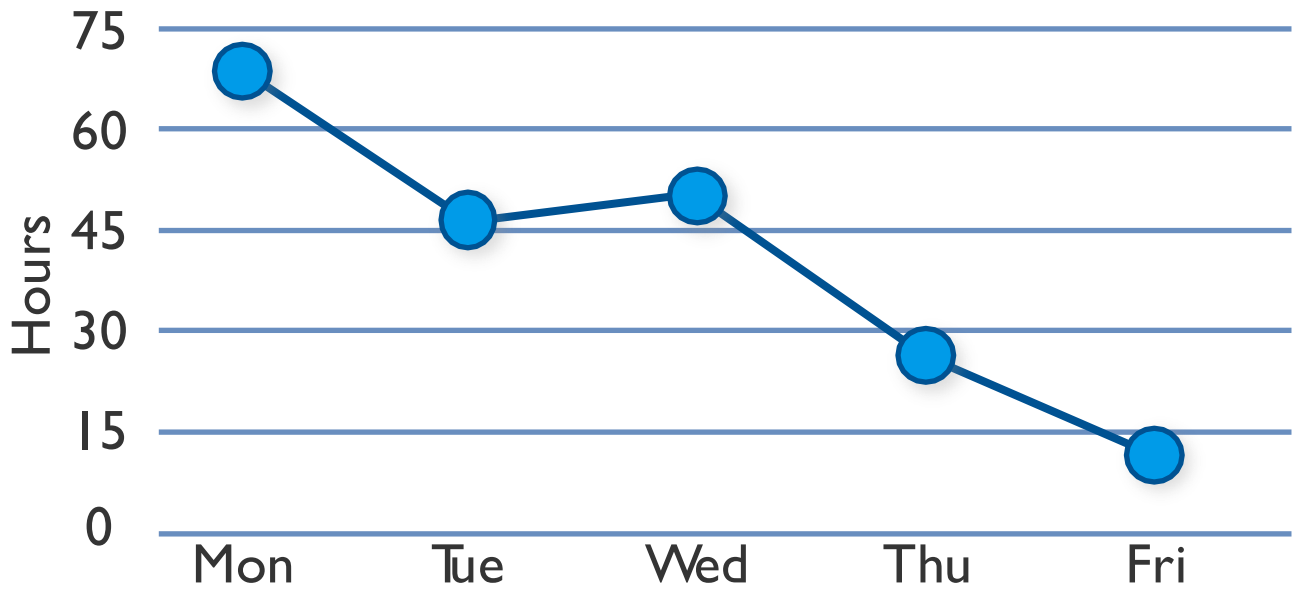
- Instead of Use Cases, Agile project owners do "user stories"
  - **Who** (user role)
  - **What** (goal)
  - **Why** (reason)
- As a [user role], I want to [goal], so I can [reason]. Example:
  - "As a customer, I want to know the monthly repayment, so that I can assess the affordability."
- **Story points**: Rating
  - common scales: 1-10 (good for capturing relative complexity)

Sample Product Backlog

Backlog item	Estimate (Story Points or hours)
Allow a customer to apply for a loan	3
As a customer, I want to know the loan repayment.	5
As a customer, I want to change the loan period.	3
As a manager, I can vary the interest rate	8
Improve interface	8
…	
…	

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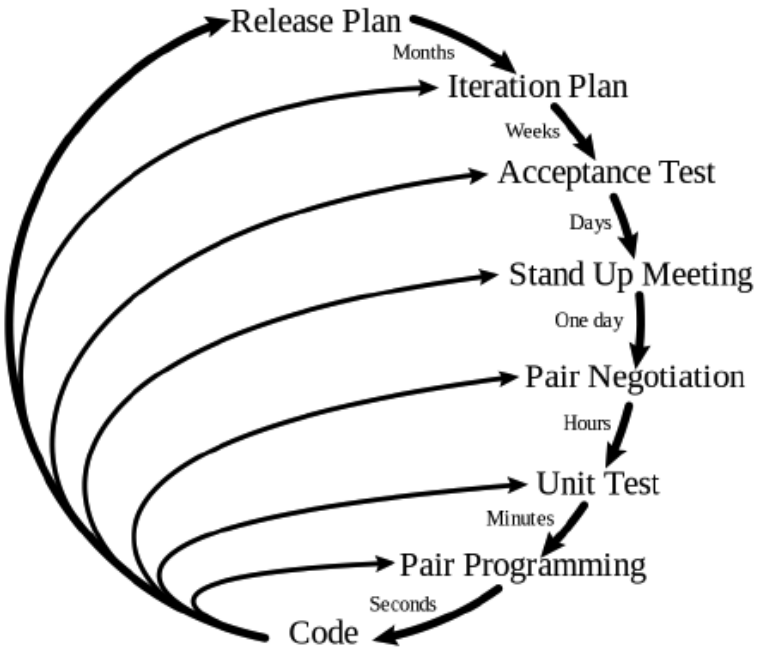
Tasks	Mon	Tue	Wed	Thu	Fri
Code the front tier	14	8	4	2	
Code the business logic	18	20	12	8	2
Test the front tier	8	10	8	8	4



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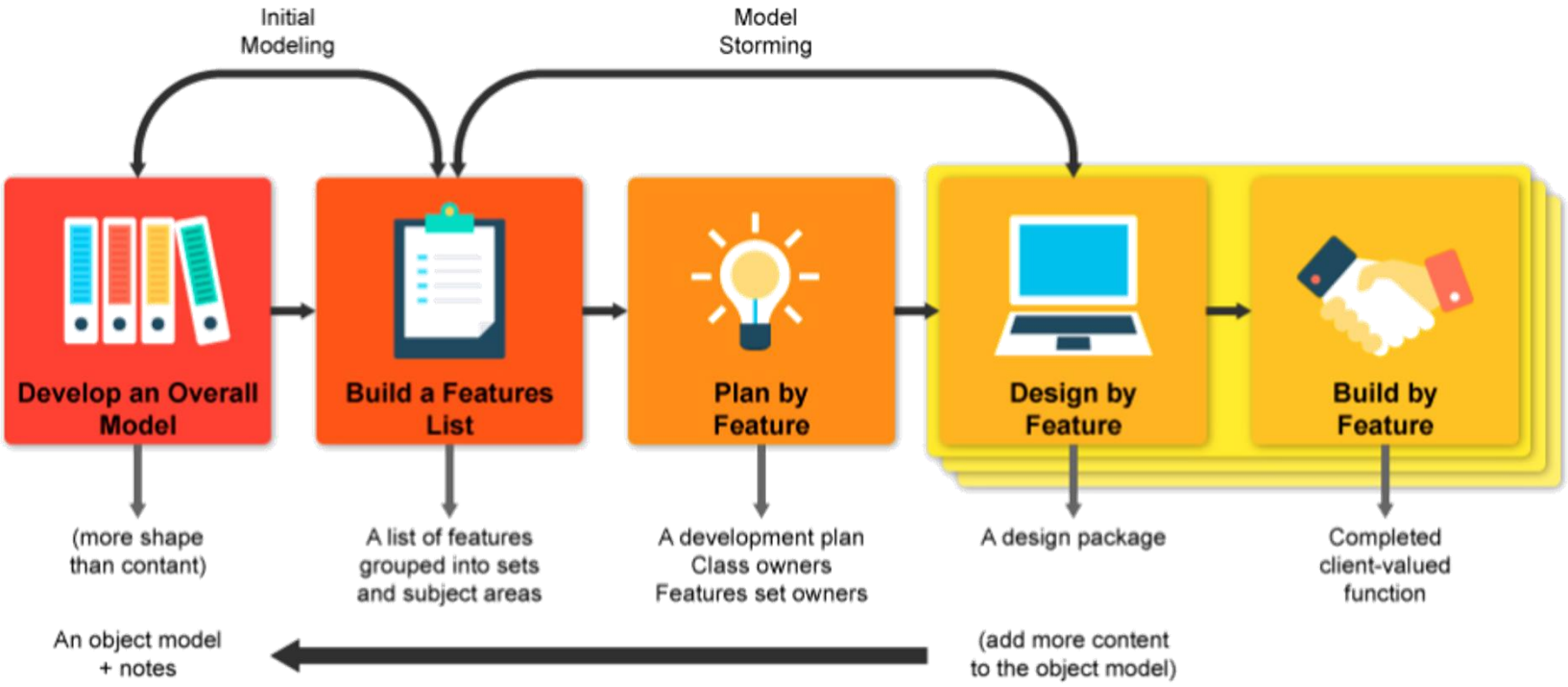
2) eXtreme Programming

A system of practices that a community of software developers is evolving to address the problems of quickly delivering quality software, and then evolving it to meet changing business needs.

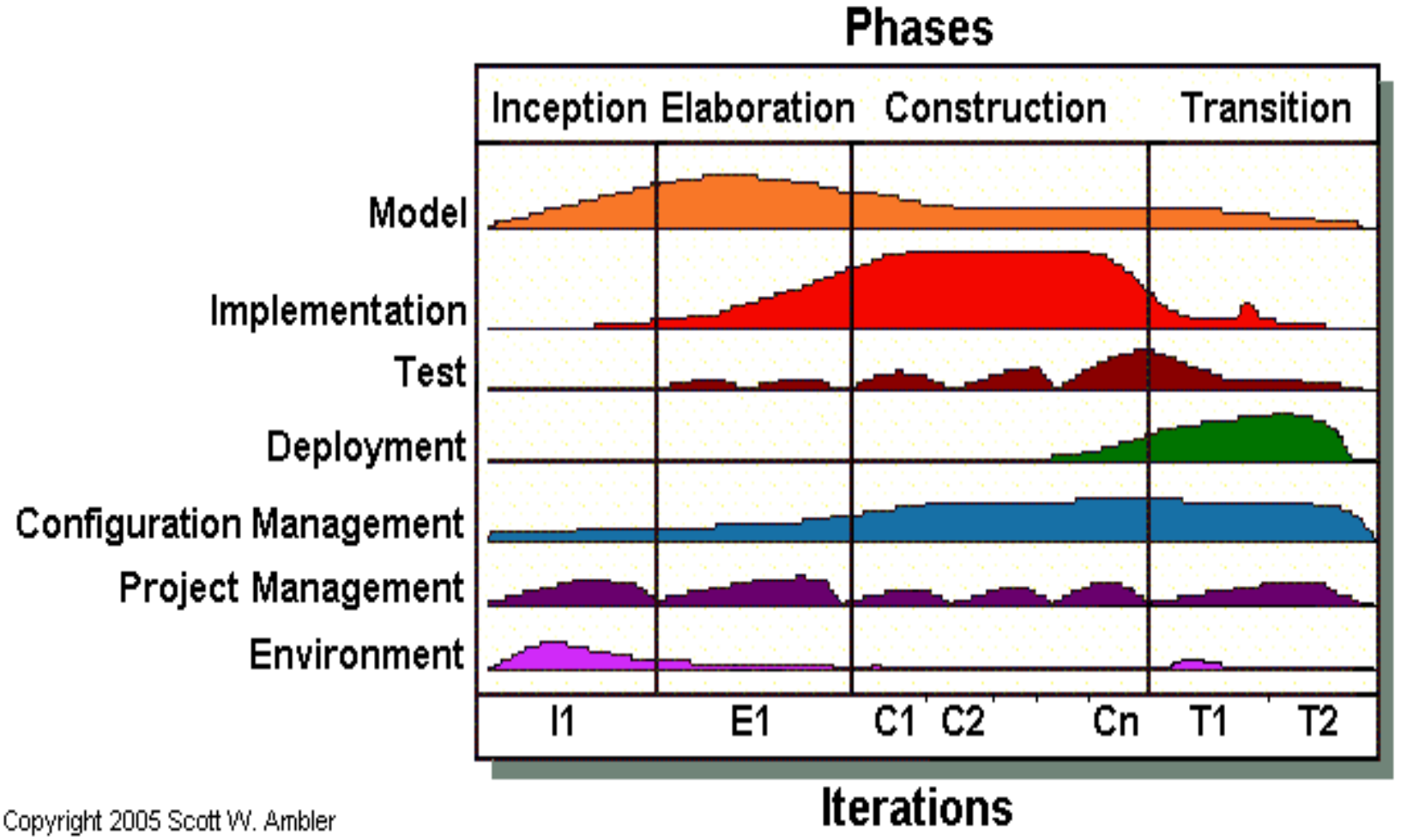




3) Feature Driven Development



4) Agile unified process



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- http://www.youtube.com/watch?v=vmGMpME_phg&feature=related

Poker Game

Using the planning poker estimate the given user story.

Discuss with your team why you game such story point.

User Story 1:

As a customer, I want to customize my product order with live 3D previews so I can visualize my purchase before checkout.

Breakout Room

Epic user story: As a healthcare provider, I want to create an electronic health record system, so I can securely manage and access patient medical data, improving patient care and reducing administrative overhead.

User Story 1: As a healthcare provider, I want to register new patients in the electronic health record system, capturing essential information such as name, date of birth, contact details, and insurance information, to create a patient profile.

Work with your group and provide at least two more user stories from the given epic user story.

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